

Mathematics for Computing

Assignment

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Definition of Statistics

Definition:

Statistics is the branch of mathematics concerned with collecting, organizing, analyzing, interpreting, and presenting data in order to draw meaningful conclusions and support decision-making.

- It helps convert raw data into useful information.
- Broadly divided into two types:
 - **Descriptive Statistics** – summarizing and describing data.
 - **Inferential Statistics** – drawing conclusions/predictions from data.

Functions and Activities:

- **Agricultural Censuses:** Tracks crop yields, land usage, and cultivation costs.
- **Economic Estimations:** Calculates GSDP, industrial growth, and inflation indices.
- **Vital Statistics:** Registers births, deaths, and demographic trends.
- **Socioeconomic Surveys:** Evaluates employment and living standards.

Functions and Activities:

- **Macroeconomic Data:** Computes national GDP, inflation indices, and industrial production.
- **National Surveys:** Measures employment, consumption, and household socio-economic metrics.
- **Demographic Tracking:** Conducts the national census and registers vital statistics.
- **Coordination:** Aligns statistical standards across ministries.

Data, Information, Knowledge, Intelligence

Data:

Raw, unprocessed facts or figures with no context.

Example: 28, 30, 32 (just numbers)

Information:

Data that has been processed/organized to give it meaning.

Example: "Temperature today is 32°C"

Knowledge:

Understanding gained by connecting information with experience.

Example: Knowing that 32°C feels hot and one should drink more water.

Intelligence:

The ability to apply knowledge to make decisions or solve new problems.

Example: Deciding to reschedule an outdoor event because of the heat.

Definition of Artificial Intelligence (AI)

Definition:

Artificial Intelligence (AI) is the branch of computer science that enables machines to simulate human intelligence – such as learning, reasoning, and problem-solving – to perform tasks and make decisions.

Example: A recommendation system that suggests movies based on past viewing behavior.

Types of Data: Univariate, Bivariate, Multivariate

Univariate Data:

Data involving **a single variable**; describes one characteristic at a time, no relationships studied. *Example:* Marks of 5 students: {45, 67, 78, 89, 90}

Bivariate Data:

Data involving **two variables**, studied together to find a relationship between them. *Example:* Height & Weight: (150, 45), (160, 55), (170, 65)

Multivariate Data:

Data involving **more than two variables**, analyzed together. *Example:* (Age, Height, Weight, Marks) = (20, 170, 65, 88)

How to Represent Data

Definition:

Data representation refers to the methods used to organize and display data so it can be understood and analyzed easily.

Common methods:

- **Tables** – rows and columns of values.
- **Bar Graph** – comparing categories.
- **Histogram** – frequency distribution of continuous data.
- **Pie Chart** – proportion of a whole.
- **Line Graph** – trend over time.

Example: Marks {45, 67, 78, 89, 90} can be shown as a bar graph with student names on the x-axis and marks on the y-axis.

Measures of Central Tendency

Definition:

A single value that represents the **center** or typical value of a dataset.

Mean: Arithmetic average of all values.

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

Example: Data 2, 4, 6, 8 $\Rightarrow \bar{x} = \frac{2+4+6+8}{4} = 5$

Median: Middle value of a dataset arranged in ascending order. *Example:*

2, 4, 6, 8, 10 $\Rightarrow$ Median = 6. For even count, 2, 4, 6, 8 $\Rightarrow \frac{4+6}{2} = 5$

Mode: Value that occurs **most frequently**. *Example:* 2, 4, 4, 6, 8 $\Rightarrow$

Mode = 4

Measures of Dispersion

Definition:

Describes **how spread out** the data values are from the central value (mean).

Range: $\text{Range} = x_{\max} - x_{\min}$ *Example:* $2, 4, 6, 8 \Rightarrow 8 - 2 = 6$

Variance: Average of the squared deviations from the mean.

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

Example: Data 2, 4, 6, 8, $\bar{x} = 5 \Rightarrow \sigma^2 = \frac{9+1+1+9}{4} = 5$

Standard Deviation: Square root of variance; same units as the data.

$$\sigma = \sqrt{\sigma^2}$$

Example: $\sigma^2 = 5 \Rightarrow \sigma = \sqrt{5} \approx 2.24$

Overview

Descriptive Statistics:

The branch of statistics that summarizes and visualizes data using quantities (mean, variance, histograms, etc.) computed directly from the data, without any probabilistic assumptions.

Frequentist Statistics:

An approach that models the data as random realizations from an unknown but **fixed** distribution, allowing us to analyze estimators and quantify their accuracy.

Topics – Descriptive: Histogram, Sample mean & variance, Order statistics, Sample covariance, Sample covariance matrix.

Topics – Frequentist: IID sampling, Mean square error, Consistency, Confidence intervals, Parametric & Non-parametric model estimation.

Histogram

Definition:

A **histogram** is obtained by binning the range of one-dimensional data and counting the number of data points that fall in each bin. It is a piecewise-constant approximation of the underlying pmf/pdf.

Key Point:

Histograms visualize the **shape/spread** of one-dimensional data; bin width controls resolution.

Sample Mean

Definition & Formula:

For data $\{x_1, \dots, x_n\}$, the sample mean is

$$\text{av}(x_1, \dots, x_n) := \frac{1}{n} \sum_{i=1}^n x_i$$

For d -dimensional vectors $\{\vec{x}_1, \dots, \vec{x}_n\}$:

$$\text{av}(\vec{x}_1, \dots, \vec{x}_n) := \frac{1}{n} \sum_{i=1}^n \vec{x}_i$$

Key Point:

Sample mean = **center of mass** of the data. **Centering** = subtracting the mean from every point so the new mean is 0.

Sample Variance & Standard Deviation

Definition & Formula:

$$\text{var}(x_1, \dots, x_n) := \frac{1}{n-1} \sum_{i=1}^n (x_i - \text{av}(x_1, \dots, x_n))^2$$

$$\text{std}(x_1, \dots, x_n) := \sqrt{\text{var}(x_1, \dots, x_n)}$$

Key Point:

Measures **average spread** around the mean. Divides by $(n-1)$, not n , so that the sample variance is an **unbiased** estimator of the true variance.

Order Statistics: Quantiles & Percentiles

Definition & Formula:

Let $x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)}$ be the sorted data. The q -quantile ($0 < q < 1$) is

$$x_{([q(n+1)])}$$

The $100p$ quantile is called the p -th percentile.

0.25 quantile = 1st quartile, 0.5 quantile = **median**, 0.75 quantile = 3rd quartile.

IQR = (3rd quartile) – (1st quartile).

Key Point:

Order statistics (median, IQR, box plots) describe a data set **more robustly** than mean/std when extreme values (outliers) are present.

Sample Covariance

Definition & Formula:

For paired data $\{(x_1, y_1), \dots, (x_n, y_n)\}$:

$$\text{cov}((x_1, y_1), \dots, (x_n, y_n)) := \frac{1}{n-1} \sum_{i=1}^n (x_i - \text{av}(x))(y_i - \text{av}(y))$$

Correlation coefficient:

$$\rho = \frac{\text{cov}(x, y)}{\text{std}(x) \text{std}(y)}, \quad -1 \leq \rho \leq 1$$

Key Point:

Covariance shows whether two features increase/decrease **together**.

Correlation is the **scale-free** (unit-free) version, always between -1 and 1 .

Sample Covariance Matrix

Definition & Formula:

For d -dimensional data $\{\vec{x}_1, \dots, \vec{x}_n\}$:

$$\Sigma(\vec{x}_1, \dots, \vec{x}_n) := \frac{1}{n-1} \sum_{i=1}^n (\vec{x}_i - \text{av})(\vec{x}_i - \text{av})^T$$

Diagonal entries = variances; off-diagonal entries = covariances between feature pairs.

Key Point:

Eigenvectors of Σ give the directions of **maximum variation** of the data – this is the basis of **Principal Component Analysis (PCA)**.

IID Sampling

Definition:

Data $x_1, x_2, \dots, x_n$ are **independent and identically distributed (iid)** if each point is drawn independently from the **same** underlying distribution. Sampling **with replacement** from a population produces iid data.

Key Point:

The iid assumption lets us treat data as realizations of random variables so probability theory can be applied to statistics.

Mean Square Error (MSE)

Definition & Formula:

For an estimator Y approximating true value $\gamma \in \mathbb{R}$:

$$\text{MSE}(Y) := E((Y - \gamma)^2)$$

Bias–Variance decomposition:

$$\text{MSE}(Y) = \underbrace{E((Y - E(Y))^2)}_{\text{variance}} + \underbrace{(E(Y) - \gamma)^2}_{\text{bias}}$$

An estimator is **unbiased** if $E(Y) = \gamma$.

Key Point:

MSE measures overall estimator error = variance + bias². Sample mean and sample variance are unbiased estimators.

Consistency

Definition:

An estimator $\tilde{Y}^{(n)}$ is **consistent** if it converges to the true value γ as $n \rightarrow \infty$ (in mean square / probability / almost surely). *Example:* The sample mean is a consistent estimator of the true mean (Law of Large Numbers). The sample median is consistent even when the mean is not well defined.

Key Point:

Consistency = “more data $\Rightarrow$ better estimate, converging to the truth.”

Confidence Intervals

Definition & Formula:

A $1 - \alpha$ confidence interval I for $\gamma \in \mathbb{R}$ satisfies

$$P(\gamma \in I) \geq 1 - \alpha, \quad 0 < \alpha < 1$$

For the mean of an iid sequence with variance $\sigma^2 \leq b^2$:

$$I_n = \left[\bar{Y}_n - \frac{b}{\sqrt{\alpha n}}, \bar{Y}_n + \frac{b}{\sqrt{\alpha n}} \right]$$

Key Point:

A confidence interval gives a **range**, not a single number, so we can quantify how sure we are about an estimate using finite data.

Non-parametric Model Estimation

Empirical CDF:

$$\hat{F}_n(x) := \frac{1}{n} \sum_{i=1}^n \mathbb{I}_{x_i \leq x}$$

Fraction of samples $\leq x$; unbiased and consistent estimator of the true CDF.

Kernel Density Estimator (bandwidth h):

$$\hat{f}_{h,n}(x) := \frac{1}{nh} \sum_{i=1}^n k\left(\frac{x - x_i}{h}\right)$$

Key Point:

No assumption about the distribution's family is made – the shape is estimated **directly from data**.

Parametric Model Estimation

Method of Moments:

Set sample moments (mean, variance, ...) equal to the theoretical moments of the assumed distribution, then solve for the parameters.

Maximum Likelihood Estimation (MLE):

Likelihood function: $L_{x_1, \dots, x_n}(\vec{\theta}) := p_{\vec{\theta}}(x_1, \dots, x_n)$ (or pdf $f_{\vec{\theta}}$ for continuous data).

$$\hat{\theta}_{ML} = \arg \max_{\vec{\theta}} \log L_{x_1, \dots, x_n}(\vec{\theta})$$

Key Point:

Assumes the data comes from a **known family** of distributions (e.g. Gaussian); the task reduces to estimating its **parameters** from data.